

Variational Autoencoder

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January 19, 2026

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1 Background

1.1 Concepts of autoencoder and variational autoencoder

An autoencoder (AE) is an unsupervised learning model that can automatically learn data features from a large number of samples, acting as a dimensionality reduction method, as shown in Figure 1 [1]. A variational autoencoder (VAE) is similar to an AE, but represents the feature as a probability distribution consisting of means and covariance matrices Figure 2. Since the feature maps are defined as a probability distribution, VAE may be trained for improved generalization and variations of outputs [2].

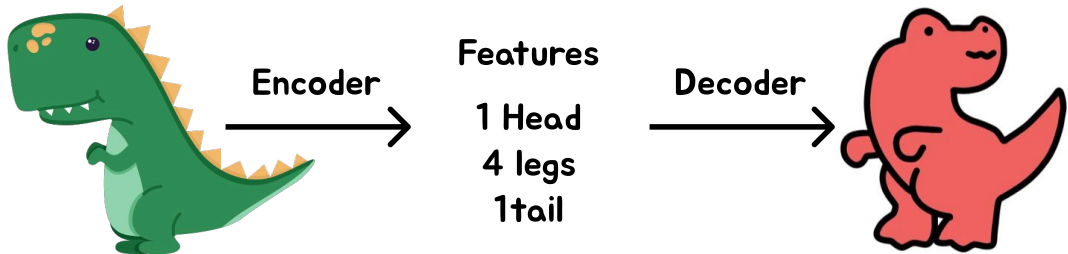


Figure 1: Concept of AE

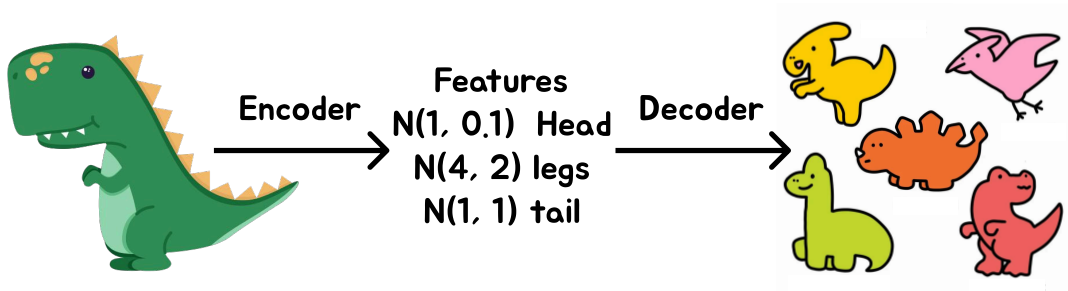


Figure 2: Concept of VAE

1.2 Probability distribution

1.2.1 1-dimension probability distribution

1-dimension probability distribution is defined as Equation 1. The mean determines translation, and standard deviation sets the width of the distribution in Figure 3.

$$p(x; \mu, \sigma) = \mathcal{N}(x; \mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right) \quad (1)$$

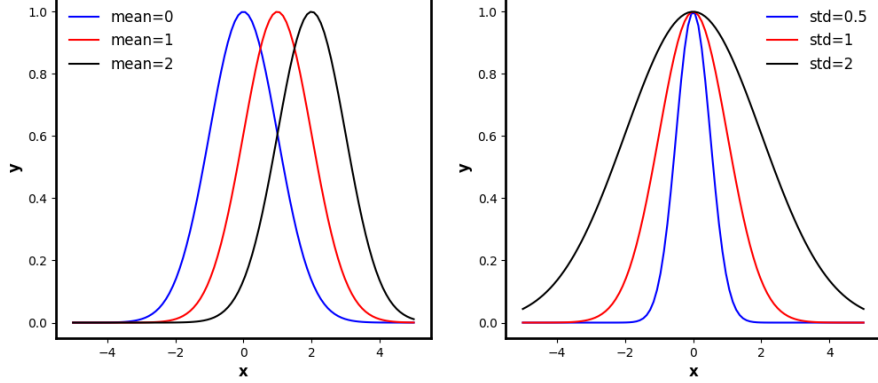


Figure 3: 1-dimensional probability distributions with various means and standard deviations

Then, the mean and standard deviation can be derived from data in Equation 2. Suppose that each data point, x_i is generated independently from probability distribution $p(x; \theta)$, where $\theta = \{\mu, \sigma\}$. The probability density from data is defined as a multiple of each probability of the data, referred to as the likelihood function for θ in Equation 3. The likelihood function is usually represented in a log format in Equation 4. To estimate the optimal θ , the optimization problem should be solved.

$$\mathcal{D} = \{x_1, x_2, \dots, x_N\} \quad (2)$$

$$p(\mathcal{D}; \theta) = p(x_1; \theta)p(x_2; \theta) \cdots p(x_N; \theta) = \prod_{i=1}^N p(x_i; \theta) \quad (3)$$

$$L(\mu, \sigma) = \log(p(\mathcal{D}; \theta)) \quad \hat{\mu}, \hat{\sigma} = \arg \max_{\mu, \sigma} L(\mu, \sigma) \quad (4)$$

based on the Equation 5, μ and σ is defined as Equation 6 and Equation 7.

$$\begin{aligned} L(\mu, \sigma) &= \log(p(\mathcal{D}; \theta)) = \log\left(\prod_{i=1}^N \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)\right) \\ &= \log\left(\prod_{i=1}^N \frac{1}{\sqrt{2\pi\sigma^2}}\right) + \log\left(\prod_{i=1}^N \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)\right) \\ &= \log\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^N + \sum_{i=1}^N \left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right) \\ &= -\frac{N}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^N (x_i - \mu)^2 \end{aligned} \quad (5)$$

$$\begin{aligned}
\frac{\partial L}{\partial \mu} = 0 &= \frac{1}{\sigma^2} \sum_{i=1}^N (x_i - \mu) \Leftrightarrow \sum_{i=1}^N (x_i - \mu) = 0 \Leftrightarrow \sum_{i=1}^N \mu = \sum_{i=1}^N x_i \\
&\Leftrightarrow N\mu = \sum_{i=1}^N x_i \quad \therefore \hat{\mu} = \frac{1}{N} \sum_{i=1}^N x_i
\end{aligned} \tag{6}$$

$$\begin{aligned}
\frac{\partial L}{\partial \sigma} = 0 &= -\frac{N}{\sigma} + \sigma^{-3} \sum_{i=1}^N (x_i - \hat{\mu})^2 \Leftrightarrow \frac{N}{\sigma} = \sigma^{-3} \sum_{i=1}^N (x_i - \hat{\mu})^2 \\
&\Leftrightarrow \sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \hat{\mu})^2 \quad \therefore \hat{\sigma} = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \hat{\mu})^2}
\end{aligned} \tag{7}$$

1.2.2 2- and multi-dimensional probability distribution

A 2-dimensional probability distribution is defined with a mean vector and a covariance matrix as Equation 8. The component in the covariance matrix is defined as Equation 9. Positive covariance means a positive proportional relationship between x_i, x_j , negative covariance means a negative proportional relationship, and zero covariance means independent probability variables Figure 4. For the multi-dimensional scale, probability distribution for probability variable $\mathbf{x} \in \mathbb{R}^D$ is represented as Equation 10

$$p(\mathbf{x}; \boldsymbol{\mu}, \Sigma) = \mathcal{N}\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}; \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \Sigma\right) = \frac{\exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})\right)}{\sqrt{(2\pi)^2 |\Sigma|}}, \Sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{12} & \sigma_{22} \end{bmatrix} \in \mathbb{R}^{2 \times 2}. \tag{8}$$

$$\Sigma_{ij} = \mathbb{E}[(x_i - \mu_i)(x_j - \mu_j)] = \frac{1}{N} \sum_{n=1}^N (x_i^{(n)} - \mu_i) (x_j^{(n)} - \mu_j) \tag{9}$$

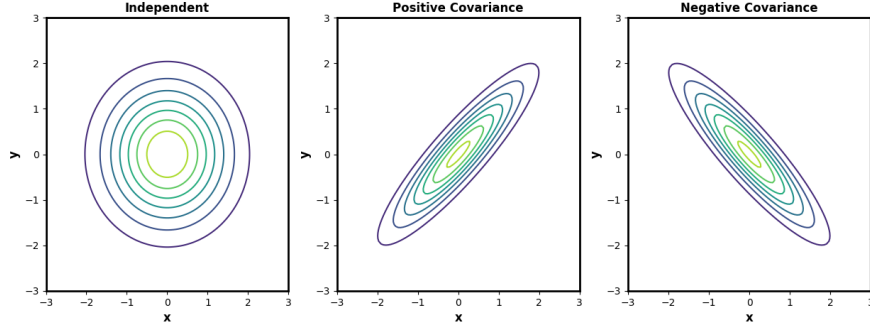


Figure 4: 2-dimensional probability distributions with various covariances

$$p(\mathbf{x}; \boldsymbol{\mu}, \Sigma) = \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \Sigma) = \frac{\exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})\right)}{\sqrt{(2\pi)^D |\Sigma|}}, \quad \mathbf{x} \in \mathbb{R}^D, \boldsymbol{\mu} \in \mathbb{R}^D, \Sigma \in \mathbb{R}^{D \times D}. \tag{10}$$

The next step is to optimize the parameter, $\theta = (\boldsymbol{\mu}, \Sigma)$, based on the data in Equation 11, defined as a likelihood function in Equation 12.

$$\mathcal{D} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N\} \quad \mathbf{x} \in \mathbb{R}^D \tag{11}$$

$$\begin{aligned}
L(\theta) &= \log \left(\prod_{i=1}^N \mathcal{N}(\mathbf{x}_i; \boldsymbol{\mu}, \Sigma) \right) = \sum_{i=1}^N \log \left(\frac{\exp(-\frac{1}{2}(\mathbf{x}_i - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x}_i - \boldsymbol{\mu}))}{\sqrt{(2\pi)^D |\Sigma|}} \right) \\
&= \sum_{i=1}^N \log \left(\frac{1}{\sqrt{(2\pi)^D |\Sigma|}} \right) + \sum_{i=1}^N \log \left(\exp \left(-\frac{1}{2}(\mathbf{x}_i - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x}_i - \boldsymbol{\mu}) \right) \right) \\
&= -\frac{N}{2} \log(2\pi)^D - \frac{N}{2} \log(|\Sigma|) - \frac{1}{2} \sum_{i=1}^N ((\mathbf{x}_i - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x}_i - \boldsymbol{\mu}))
\end{aligned} \tag{12}$$

The mean vector can be defined as Equation 13.

$$\begin{aligned}
\frac{\partial L(\theta)}{\partial \boldsymbol{\mu}} = 0 &= \frac{\partial}{\partial \boldsymbol{\mu}} \left(-\frac{1}{2} \sum_{i=1}^N ((\mathbf{x}_i - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x}_i - \boldsymbol{\mu})) \right) \\
&= -\frac{1}{2} \sum_{i=1}^N (-2|\Sigma^{-1}|(\mathbf{x}_i - \boldsymbol{\mu})) = |\Sigma^{-1}| \sum_{i=1}^N (\mathbf{x}_i - \boldsymbol{\mu}) = |\Sigma^{-1}| \sum_{i=1}^N \mathbf{x}_i - N\boldsymbol{\mu} \\
\therefore \hat{\boldsymbol{\mu}} &= \frac{1}{N} \sum_{i=1}^N \mathbf{x}_i
\end{aligned} \tag{13}$$

The covariance matrix can be defined as Equation 14.

$$\begin{aligned}
\frac{\partial L(\theta)}{\partial \Sigma} = 0 &= \frac{\partial}{\partial \Sigma} \left(-\frac{N}{2} \log(|\Sigma|) - \frac{1}{2} \sum_{i=1}^N ((\mathbf{x}_i - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x}_i - \boldsymbol{\mu})) \right) \\
&= -\frac{N}{2} (\Sigma^{-1})^T + \frac{1}{2} (\Sigma^{-1} S \Sigma^{-1})^T, \quad S = \sum_{i=1}^N ((\mathbf{x}_i - \boldsymbol{\mu})(\mathbf{x}_i - \boldsymbol{\mu})^T) \\
&\Leftrightarrow N(\Sigma^{-1})^T = (\Sigma^{-1} S \Sigma^{-1})^T \Leftrightarrow \Sigma = \frac{1}{N} S \\
\therefore \hat{\Sigma} &= \frac{1}{N} \sum_{i=1}^N ((\mathbf{x}_i - \hat{\boldsymbol{\mu}})(\mathbf{x}_i - \hat{\boldsymbol{\mu}})^T)
\end{aligned} \tag{14}$$

1.2.3 Gaussian mixture model

The Gaussian mixture model (GMM) means the mixture of multiple normal distributions in Equation 15 as Figure 5.

$$p(\mathbf{x}; \theta) = 0.5\mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_1, \Sigma_1) + 0.5\mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_2, \Sigma_2) \quad (15)$$

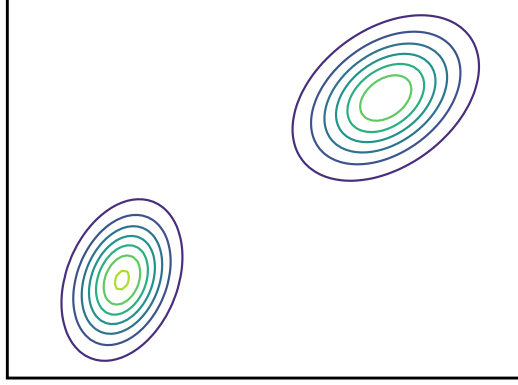


Figure 5: Example of GMM

In a more general format, the probability distribution consists of K normal distributions represented with a mean matrix, covariance matrices, and ratio of each normal distribution, ϕ_k , for input data $\mathbf{x} \in \mathbb{R}^D$ in Equation 16.

$$\begin{aligned} \boldsymbol{\mu} &= \{\boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \dots, \boldsymbol{\mu}_K\} \quad \boldsymbol{\mu}_k \in \mathbb{R}^D \\ \boldsymbol{\Sigma} &= \{\boldsymbol{\Sigma}_1, \boldsymbol{\Sigma}_2, \dots, \boldsymbol{\Sigma}_K\} \quad \boldsymbol{\Sigma}_k \in \mathbb{R}^{D \times D} \\ p(\mathbf{x}) &= \phi_1\mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1) + \phi_2\mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_2, \boldsymbol{\Sigma}_2) + \dots + \phi_K\mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_K, \boldsymbol{\Sigma}_K) \\ &= \sum_{k=1}^K \phi_k\mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k), \quad \sum_{k=1}^K \phi_k = 1 \end{aligned} \quad (16)$$

To estimate the parameter from data, $\theta = \{\phi_k, \boldsymbol{\mu}, \boldsymbol{\Sigma}\}$, the log likelihood function is expressed as Equation 17. However, the log-sum format is difficult to solve using numerical analysis methods. Therefore, different approaches should be introduced.

$$\begin{aligned} L(\theta) &= \log(p(\mathcal{D}; \theta)) = \log \left(\prod_{i=1}^N p(\mathbf{x}_i; \theta) \right) \\ &= \sum_{i=1}^N \log(p(\mathbf{x}_i; \theta)) \\ &= \sum_{i=1}^N \log \left(\sum_{k=1}^K (\phi_k \mathcal{N}(\mathbf{x}_i; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)) \right) \end{aligned} \quad (17)$$

1.3 Expectation maximization algorithm

The expectation notation for probability distribution, $p(x)$, and function, $f(x)$, is expressed as Equation 18.

$$\mathbb{E}_{p(x)}[f(x)] = \sum_x p(x)f(x) \quad (18)$$

1.3.1 KL divergence

Kullback-Leibler divergence (KL divergence) is a measure of the difference between two probability distributions. The KL divergence for the target probability distribution, $p(x)$, for the estimated probability distribution, $q(x)$, is defined as Equation 19 [3]. In more detail, KL divergence consists of the cross-entropy of $p(x)$ with respect to $q(x)$ and the entropy of $p(x)$. The value of KL divergence is a non-negative value, and zero when $p(x)$ is exactly equal to $q(x)$.

$$\begin{aligned} D_{KL}(p(x)||q(x)) &= \sum_x p(x) \log \left(\frac{p(x)}{q(x)} \right) \\ &= - \sum_x p(x) \log(q(x)) + \sum_x p(x) \log(p(x)) \\ &= -\mathbb{E}_{p(x)}[\log(q(x))] + \mathbb{E}_{p(x)}[\log(p(x))] \end{aligned} \quad (19)$$

The KL divergence is represented in the log likelihood function of GMM in Equation 17. Suppose that the target probability distribution, $p_*(x)$, data from $p_*(x)$, $\mathcal{D} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N\}$ and estimated probability distribution, $p_\theta(x)$. Since the data, $\mathcal{D}$ follows the target probability distribution, expectation for $\mathbb{E}_{p_*(x)}[\log(q(x))]$ can be expressed as $\frac{1}{N} \sum_{i=1}^N \log(q(x_i))$ in Equation 21. The objective is minimize the KL divergence for θ , objective function of KL divergence can be expressed as ??

$$\begin{aligned} D_{KL}(p_*(x)||p_\theta(x)) &= \sum_{i=1}^N p_*(x) \log \left(\frac{p_*(x_i)}{p_\theta(x_i)} \right) \\ &\approx \frac{1}{N} \sum_{i=1}^N \log \left(\frac{p_*(x_i)}{p_\theta(x_i)} \right) \\ &= \frac{1}{N} \sum_{i=1}^N (\log(p_*(x_i)) + \log(p_\theta(x_i))) \end{aligned} \quad (20)$$

$$\begin{aligned} \arg \min_{\theta} D_{KL}(p_*(x)||p_\theta(x)) &= \arg \min_{\theta} \left(\frac{1}{N} \sum_{i=1}^N (\log(p_*(x_i)) + \log(p_\theta(x_i))) \right) \\ &\approx \arg \min_{\theta} \left(- \sum_{i=1}^N \log(p_\theta(x_i)) \right) \\ &= \arg \max_{\theta} \sum_{i=1}^N (\log(p_\theta(x_i))) \\ &= \arg \max_{\theta} L(\theta) \end{aligned} \quad (21)$$

1.3.2 Model with latent space

With K latent variables and data $\mathcal{D}$, the probability for data $\mathbf{x}_i$ is expressed as a conditional probability in Equation 22. To deal with the intractable conditional probability $p(z_i | \mathbf{x}_i)$, we introduce an auxiliary distribution $q(z_i)$. The distribution $q(z_i)$ represents a probability distribution over the latent variable z_i associated with the data point $\mathbf{x}_i$, and is used to approximate the true posterior $p(z_i | \mathbf{x}_i)$ in Equation 24.

$$\begin{aligned}
p_\theta(\mathbf{x}_i) &= \sum_{j=1}^K p_\theta(\mathbf{x}_i, z_i = j) \\
&= \sum_{j=1}^K \phi_j \mathcal{N}(\mathbf{x}_i; \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j) \\
&= \frac{p_\theta(\mathbf{x}_i, z_i = k)}{p_\theta(z_i = k | \mathbf{x}_i)}
\end{aligned} \tag{22}$$

$$q(z_i = k) \approx p_\theta(z_i = k | \mathbf{x}_i) \tag{23}$$

log likelihood function for data, $\mathbf{x}_i$ is represented with latent variable as Equation 24.

$$\begin{aligned}
L(\theta) &= \log(p_\theta(\mathbf{x}_i)) = \log\left(\frac{p_\theta(\mathbf{x}_i, z_i = k)}{p_\theta(z_i = k | \mathbf{x}_i)}\right) \\
&= \log\left(\frac{p_\theta(\mathbf{x}_i, z_i = k)}{p_\theta(z_i = k | \mathbf{x}_i)} \frac{q(z_i = k)}{q(z_i = k)}\right) \\
&= \log\left(\frac{p_\theta(\mathbf{x}_i, z_i = k)}{q(z_i = k)}\right) + \log\left(\frac{q(z_i = k)}{p_\theta(z_i = k | \mathbf{x}_i)}\right) \\
&= \sum_{j=1}^K q(z_i = j) \left(\log\left(\frac{p_\theta(\mathbf{x}_i, z_i = j)}{q(z_i = j)}\right) + \log\left(\frac{q(z_i = j)}{p_\theta(z_i = j | \mathbf{x}_i)}\right) \right), \quad \sum_{j=1}^K q(z_i = j) = 1 \\
&= D_{KL}(q(z_i) || p_\theta(z_i | \mathbf{x}_i)) + \sum_{j=1}^K q(z_i = j) \log\left(\frac{p_\theta(\mathbf{x}_i, z_i = j)}{q(z_i = j)}\right), \quad D_{KL} \geq 0 \\
&\geq \sum_{j=1}^K q(z_i = j) \log\left(\frac{p_\theta(\mathbf{x}_i, z_i = j)}{q(z_i = j)}\right)
\end{aligned} \tag{24}$$

Then, the lower bound of the log likelihood function can be constructed, referred to as ELBO in Equation 25. log likelihood function for data, $\mathcal{D}$ is expressed as Equation 26

$$\begin{aligned}
ELBO(\mathbf{x}_i; q, \theta) &= \sum_{j=1}^K q(z_i = j) \log\left(\frac{p_\theta(\mathbf{x}_i, z_i = j)}{q(z_i = j)}\right) \\
&= \mathbb{E}_{q(z_i)} \left[\log\left(\frac{p_\theta(\mathbf{x}_i, z_i = j)}{q(z_i = j)}\right) \right]
\end{aligned} \tag{25}$$

$$L(\theta) = \log(p_\theta(\mathcal{D})) \geq \sum_{i=1}^N ELBO(\mathbf{x}_i; q(z_i), \theta) \tag{26}$$

1.3.3 E-step

The EM algorithm consists of the E-step and M-step. E-step updates $q(z_i)$, and M-step updates $\phi_k, \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k$. In the E-step, based on the fixed $\phi_k, \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k$ the $q(z_i)$ is updated as Equation 27.

$$\begin{aligned} q(z_i = k) &= p_\theta(z = k | \mathbf{x}_i) = \frac{p_\theta(z = k, \mathbf{x}_i)}{p_\theta(\mathbf{x}_i)} \\ &= \frac{\phi_k \mathcal{N}(\mathbf{x}_i; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^K \phi_j \mathcal{N}(\mathbf{x}_i; \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)} \end{aligned} \quad (27)$$

1.3.4 M-step

For the fixed $q(z_i)$, The log likelihood function is expressed as Equation 28. by extracting the term relative to $\phi_k, \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k$ the $q(z_i)$ is defined as Equation 29 So, based on the three equation, parameter can be optimized as Equation 30.

$$\begin{aligned} L(\theta) &\geq \sum_{i=1}^N ELBO(\mathbf{x}_i; q(z_i), \theta) \\ &= \sum_{i=1}^N \sum_{j=1}^K q(z_i = j) \log \left(\frac{p_\theta(\mathbf{x}_i, z_i = j)}{q(z_i = j)} \right) \\ &= \sum_{i=1}^N \sum_{j=1}^K q(z_i = j) \log(p_\theta(\mathbf{x}_i, z_i = j)) - \sum_{i=1}^N \sum_{j=1}^K q(z_i = j) \log(q(z_i = j)) \end{aligned} \quad (28)$$

$$J_1(\theta) = \sum_{i=1}^N \sum_{j=1}^K q(z_i = j) \log(p_\theta(\mathbf{x}_i, z_i = j)) = \sum_{i=1}^N \sum_{j=1}^K q(z_i = j) \log(\phi_j \mathcal{N}(\mathbf{x}_i; \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)) \quad (29)$$

$$\frac{\partial J_1}{\partial \boldsymbol{\mu}} = \frac{\partial J_1}{\partial \boldsymbol{\Sigma}} = \frac{\partial J_1}{\partial \boldsymbol{\phi}} = 0 \quad (30)$$

The mean vector for k latent space is updated as Equation 31

$$\boldsymbol{\mu}_k = \frac{\sum_{i=1}^N q(z_i = k) \mathbf{x}_i}{\sum_{i=1}^N q(z_i = k)} \quad (31)$$

The covariance matrix for k latent space is updated as Equation 32.

$$\boldsymbol{\Sigma}_k = \frac{\sum_{i=1}^N q(z_i = k) (\mathbf{x}_i - \boldsymbol{\mu}_k)(\mathbf{x}_i - \boldsymbol{\mu}_k)^T}{\sum_{i=1}^N q(z_i = k)} \quad (32)$$

The mixing coefficient for k latent space is updated as Equation 33.

$$\phi_k = \frac{1}{N} \sum_{i=1}^N q(z_i = k) \quad (33)$$

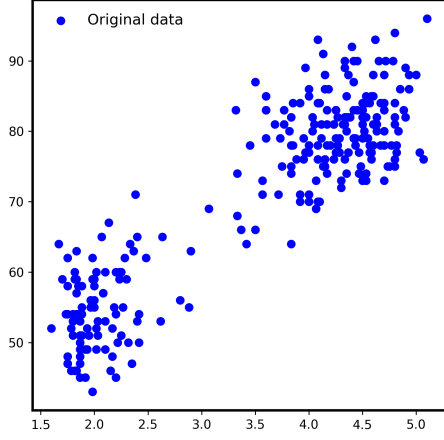


Figure 6: Original data

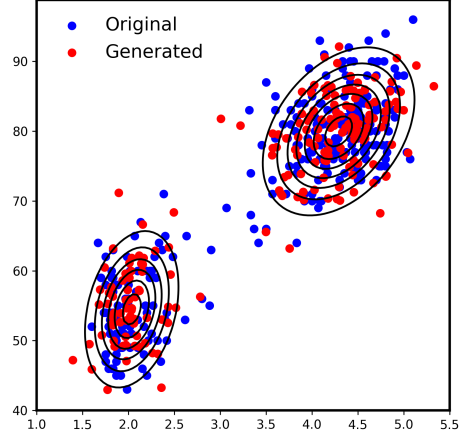


Figure 7: Original and generated data

2 VAE

2.1 Encoder and Decoder

For a unimodal latent variable, the encoder, parameterized by a neural network with parameters W_e , maps an input image to the mean and standard deviation vectors of the latent distribution, as shown in Figure 8. The encoder outputs the log-variance so that its values are unconstrained over the real numbers. The latent vector, z , is defined as Equation 34. Decoder with network parameter, W_d , transforms from latent vector to model output in Figure 9.

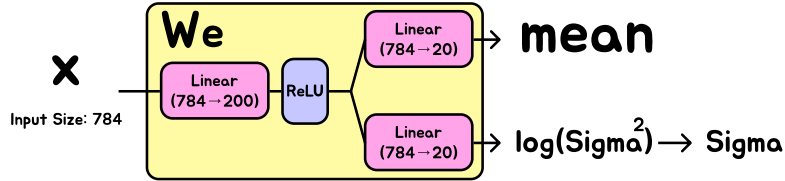


Figure 8: Concept of encoder with 200 hidden dimensions and 20 latent dimensions.

$$z = \mu + \sigma \cdot \epsilon, \quad \epsilon \approx \mathcal{N}(0, I) \quad (34)$$

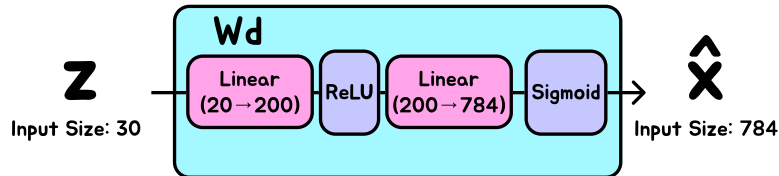


Figure 9: Concept of decoder

2.2 Loss

With K latent space, the dimension of $\boldsymbol{\mu}, \boldsymbol{\Sigma}$ is K , in Equation 35. In the covariance matrix, the correlation between different axes is zero.

$$\boldsymbol{\mu} \in \mathbb{R}^K, \quad \boldsymbol{\Sigma} = \begin{bmatrix} \sigma_1^2 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_K^2 \end{bmatrix}, \quad \sigma_{ij} = 0 (i \neq j), \quad \boldsymbol{\sigma} \in \mathbb{R}^K \quad (35)$$

The role of encoder and decoder is defined as Equation 36.

$$\begin{aligned} \boldsymbol{\mu}, \boldsymbol{\sigma} &= \text{Encoder}(\mathbf{x}; W_e) \\ \hat{\mathbf{x}} &= \text{Decoder}(z; W_d), \quad z = \boldsymbol{\mu} + \boldsymbol{\sigma} \cdot \epsilon \end{aligned} \quad (36)$$

The probability distribution of the latent variable for input, $q_{W_e}(z|x)$, and the probability distribution of input for the latent variable, p_{W_d} , are defined as Equation 37. In this equation, $p(z)$ denotes a prior distribution that serves as a criterion for $q_{W_e}(z|x)$, since random samples drawn from $p(z)$ must be decodable by the model from the decoder's perspective, ensuring that samples drawn from $p(z)$ can be meaningfully decoded..

$$\begin{aligned} q_{W_e}(z|\mathbf{x}) &= \mathcal{N}(z; \boldsymbol{\mu}, \boldsymbol{\sigma}\mathbf{I}) \\ p_{W_d}(\mathbf{x}|z) &= \mathcal{N}(\mathbf{x}; \hat{\mathbf{x}}, \mathbf{I}) \\ p(z) &= \mathcal{N}(z; \mathbf{0}, \mathbf{I}) \end{aligned} \quad (37)$$

The log likelihood function has a lower bound, referred to as ELBO, which is expressed as Equation 38.

$$\begin{aligned} L(W_e, W_d) &= \log(p_{W_e, W_d}(\mathcal{D})) \geq \sum_{i=1}^N \text{ELBO}(\mathbf{x}_i; W_e, W_d) \\ &= \sum_{i=1}^N \left(\int_{-\infty}^{\infty} q_{W_e}(z|\mathbf{x}_i) \log \left(\frac{p_{W_d}(\mathbf{x}_i, z)}{q_{W_e}(z|\mathbf{x}_i)} \right) dz \right) \end{aligned} \quad (38)$$

ELBO can be divided as expectation term and KL divergence term in Equation 39 and Equation 41.

$$\begin{aligned} \text{ELBO}(\mathbf{x}_i; W_e, W_d) &= \int q_{W_e}(z|\mathbf{x}_i) \log \left(\frac{p_{W_d}(\mathbf{x}_i, z)}{q_{W_e}(z|\mathbf{x}_i)} \right) dz \\ &= \int q_{W_e}(z|\mathbf{x}_i) \log \left(\frac{p_{W_d}(\mathbf{x}_i|z)p(z)}{q_{W_e}(z|\mathbf{x}_i)} \right) dz \\ &= \int q_{W_e}(z|\mathbf{x}_i) \log(p_{W_d}(\mathbf{x}_i|z)) dz - \int q_{W_e}(z|\mathbf{x}_i) \log \left(\frac{p(z)}{q_{W_e}(z|\mathbf{x}_i)} \right) dz \\ &= \mathbb{E}_{q_{W_e}(z|\mathbf{x}_i)}[\log(p_{W_d}(\mathbf{x}_i|z))] - D_{KL}(q_{W_e}(z|\mathbf{x}_i)||p(z)) \end{aligned} \quad (39)$$

$$J_1 = \mathbb{E}_{q_{W_e}(z|\mathbf{x}_i)}[\log(p_{W_d}(\mathbf{x}_i|z))], \quad J_2 = D_{KL}(q_{W_e}(z|\mathbf{x}_i)||p(z)) \quad (40)$$

The reconstruction error, J_1 , means the difference between the real input image and the predicted image. From model architecture perspectives, J_1 is expressed as Equation 41

$$\begin{aligned}
J_1 &= \mathbb{E}_{q_{W_e}(z|\mathbf{x}_i)}[\log(p_{W_d}(\mathbf{x}_i|z))] \approx \log(\mathcal{N}(\mathbf{x}_i; \hat{\mathbf{x}}, \mathbf{I})) \\
&= \log\left(\frac{1}{\sqrt{(2\pi)^D|\mathbf{I}|}} \exp\left(-\frac{1}{2}(\mathbf{x}_i - \hat{\mathbf{x}})^T \mathbf{I}^{-1}(\mathbf{x}_i - \hat{\mathbf{x}})\right)\right) \\
&= -\frac{1}{2}(\mathbf{x}_i - \hat{\mathbf{x}})^T(\mathbf{x}_i - \hat{\mathbf{x}}) + \log\left(\frac{1}{\sqrt{(2\pi)^D|\mathbf{I}|}}\right) \\
&= -\frac{1}{2} \sum_{d=1}^D (\mathbf{x}_{i,d} - \hat{\mathbf{x}}_d)^2 + \text{const} \\
&\approx -\frac{1}{2} \text{MSE}(\mathbf{x}_i, \hat{\mathbf{x}})
\end{aligned} \tag{41}$$

The regularization term, J_2 , constrains the discrepancy between the encoder-induced latent distribution and the prior distribution of latent dimension, H , which is measured by the KL divergence in Equation 43.

$$\begin{aligned}
J_2 &= D_{KL}(q_{W_e}(z|\mathbf{x}_i)||p(z)), \quad q_{W_e}(z|\mathbf{x}_i) = \mathcal{N}(z; \boldsymbol{\mu}, \boldsymbol{\sigma}\mathbf{I}), \quad p(z) = \mathcal{N}(z; \mathbf{0}, \mathbf{I}) \\
&= -\frac{1}{2} \sum_{h=1}^H (1 + \log \sigma_h^2 - \mu_h^2 - \sigma_h^2)
\end{aligned} \tag{42}$$

Finally, the optimization process is maximizing ELBO expressed as J_1 and J_2 as Equation 43.

$$\begin{aligned}
\arg \max_{W_e, W_d} \text{ELBO} &= \arg \max_{W_e, W_d} J_1 - J_2 \\
&\approx \arg \max_{W_e, W_d} -\frac{1}{2} \sum_{d=1}^D (\mathbf{x}_{i,d} - \hat{\mathbf{x}}_d)^2 + \frac{1}{2} \sum_{h=1}^H (1 + \log \sigma_h^2 - \mu_h^2 - \sigma_h^2) \\
&\approx \arg \min_{W_e, W_d} \text{MSE}(\mathbf{x}_i, \hat{\mathbf{x}}) + \sum_{h=1}^H (1 + \log \sigma_h^2 - \mu_h^2 - \sigma_h^2)
\end{aligned} \tag{43}$$

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